CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 11: Areas Related to Circles

Topic-wise short notes with solved examples

A sector is a "slice" of a circle and a segment is the part cut off by a chord. This chapter is about finding their areas and the length of an arc.

1 Circle: a quick recap

- For a circle of radius r: circumference $= 2\pi r$ and area $= \pi r^2$. Diameter $d = 2r$.
- Use $\pi = \dfrac{22}{7}$ unless the question gives another value such as 3.14.
- Area of a semicircle $= \dfrac{1}{2}\pi r^2$. Area of a quadrant $= \dfrac{1}{4}\pi r^2$.
- Area of a ring between two concentric circles of radii R and r is $\pi(R^2 - r^2)$.

2 Sector of a circle

- A **sector** is the region enclosed by two radii and the arc between them.
- The angle between the two radii is the **angle of the sector**, θ.
- The smaller region is the **minor sector** (angle θ) and the larger is the **major sector** (angle $360^\circ - \theta$).
- The whole circle is a sector of angle 360°, so a sector of angle θ is the fraction $\dfrac{\theta}{360}$ of the circle.

$$\text{Area of sector} = \frac{\theta}{360} \times \pi r^2 \qquad \text{Length of arc} = \frac{\theta}{360} \times 2\pi r$$

- Perimeter of a sector $=$ length of arc $+ 2r$.
- Area of the major sector $= \pi r^2 -$ area of the minor sector.

Example 2.1 Find the area of a sector of a circle of radius 7 cm with angle 60°, and the length of its arc.

SOLUTION

Area $= \dfrac{60}{360} \times \dfrac{22}{7} \times 7 \times 7 = \dfrac{1}{6} \times 154 = \dfrac{77}{3}$ cm²

Arc $= \dfrac{60}{360} \times 2 \times \dfrac{22}{7} \times 7 = \dfrac{1}{6} \times 44 = \dfrac{22}{3}$ cm

Area ≈ 25.67 cm²; arc length ≈ 7.33 cm

Example 2.2 The minute hand of a clock is 21 cm long. Find the area it sweeps in 10 minutes.

SOLUTION

In 60 minutes the minute hand turns 360°, so in 10 minutes it turns 60°.

$$\text{Area} = \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{6} \times 1386$$

Area $= 231$ cm²

Example 2.3 A horse is tied to a peg at one corner of a square field of side 15 m by a rope 7 m long. Find the area of the part of the field in which it can graze.

SOLUTION

The corner of a square is 90°, so the horse grazes a sector of radius 7 m and angle 90°.

$$\text{Area} = \frac{90}{360} \times \frac{22}{7} \times 7 \times 7 = \frac{1}{4} \times 154$$

Area $= 38.5$ m²

REMEMBER

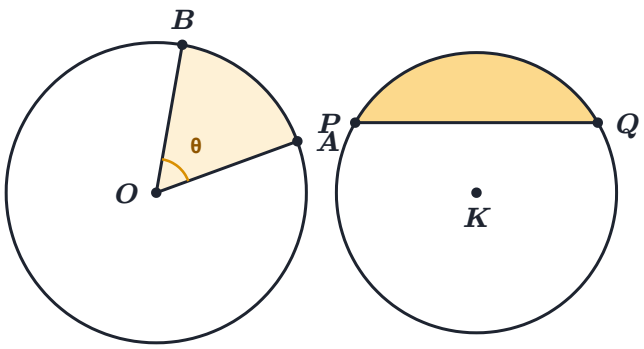
The minute hand turns 6° per minute. The hour hand turns 30° per hour, which is 0.5° per minute.

3 Segment of a circle

- A **segment** is the region between a chord and the arc it cuts off.
- The smaller region is the **minor segment** and the larger is the **major segment**.

$$\text{Area of minor segment} = \text{area of the sector} - \text{area of the triangle formed by the chord and the two radii}$$

- Area of the major segment $= \pi r^2 -$ area of the minor segment.

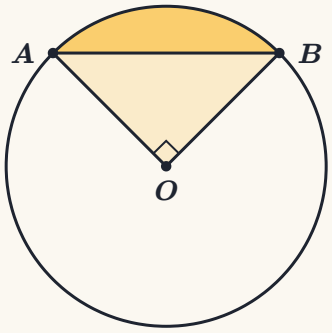


Left: sector OAB. Right: the minor segment cut off by the chord PQ.

- The triangle is isosceles with two sides equal to r and the angle θ between them.

Angle θ	Area of the triangle
60°	$\frac{\sqrt{3}}{4}r^2$ (equilateral triangle)
90°	$\frac{1}{2}r^2$ (right-angled triangle)
120°	$\frac{\sqrt{3}}{4}r^2$

Example 3.1 A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the minor segment and of the major segment.



Minor segment (yellow) = sector OAB − triangle OAB

SOLUTION

Area of sector $= \frac{90}{360} \times \frac{22}{7} \times 14 \times 14 = 154$ cm²

Area of triangle $= \frac{1}{2} \times 14 \times 14 = 98$ cm²

Minor segment $= 154 - 98 = 56$ cm²

Area of circle $= \frac{22}{7} \times 14 \times 14 = 616$ cm², so major segment $= 616 - 56 = 560$ cm²

Minor segment $= 56$ cm²; major segment $= 560$ cm²

Example 3.2 A chord of a circle of radius 21 cm subtends an angle of 60° at the centre. Find the area of the minor segment. Take $\sqrt{3} = 1.732$.

SOLUTION

Area of sector $= \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231$ cm²

The triangle is equilateral with side 21 cm: area $= \frac{\sqrt{3}}{4} \times 21 \times 21 = 110.25 \times 1.732 \approx 190.95$ cm²

Area of the minor segment $\approx 231 - 190.95 = 40.05$ cm²

4 Finding the area of the triangle for other angles

- Draw the perpendicular OM from the centre to the chord AB. It bisects the chord and the angle θ.
- $OM = r\cos\frac{\theta}{2}$ and $AM = r\sin\frac{\theta}{2}$, so the chord $AB = 2r\sin\frac{\theta}{2}$.

- Area of triangle $= \frac{1}{2} \times AB \times OM$.

Example 4.1 Find the area of the triangle formed by two radii and a chord that subtends 120° at the centre of a circle of radius 12 cm.

SOLUTION

$OM = 12\cos 60^\circ = 6$ cm and $AM = 12\sin 60^\circ = 6\sqrt{3}$ cm, so $AB = 12\sqrt{3}$ cm

Area $= \frac{1}{2} \times 12\sqrt{3} \times 6$

Area $= 36\sqrt{3} \approx 62.35$ cm²

5 Common mistakes

- Using the diameter in place of the radius.
- Forgetting to add $2r$ when the **perimeter** of a sector is asked.
- Giving the sector's area when the segment's area is asked.
- Using $\frac{22}{7}$ when the question says $\pi = 3.14$, or the other way round.
- Leaving out the units: cm for length, cm² for area.

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